

Project Report On

Real-Time EV Battery Management System with Safety Protections



Submitted in Partial Fulfilment for the award of

**Post Graduate Certificate Program in Embedded
Systems Design (PGCP-DESD)**

from
C-DAC, ACTS (Pune)

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ACKNOWLEDGEMENT

This project “Real-Time Vehicle Management System with Safety Protections, SOC estimation and IOT dashboard” was a great learning experience for us and we are submitting this work to Advanced Computing Training School (CDAC ACTS).

We are very glad to mention the name of *Mr. Arvind Kumar* for their valuable guidance to work on this project. Their guidance and support helped me to overcome various obstacles and intricacies during project work.

Our heartfelt thanks goes to *Ms. Swati Salunkhe Ma'am* (Course Coordinator, *PGCP-DESD*) who gave all the required support and kind coordination to provide all the necessities like required hardware, internet facility, and extra Lab hours to complete the project and throughout the course up to the last day here in C-DAC ACTS, Pune.

LIST OF FIGURES

Figure No.	Title
Figure 1	STM32F407VGT6
Figure 2	ESP32-WROOM-32 Module
Figure 3	INA219 Current sensor
Figure 4	NTC thermistor
Figure 5	MQ135 Smoke Sensor
Figure 6	SN65HVD230 CAN Transceiver
Figure 7	Buck Conveter Module
Figure 8	L293N Motor Driver
Figure 9	Block Digram
Figure 10	Detailed Block Digram

TABLE OF CONTENTS

❖ ABSTRACT

1. INTRODUCTION

- 1.1. Introduction
- 1.2 Problem Statement
- 1.3 Objective
- 1.4 Scope of Project
- 1.5. Motivation

2. LITERATURE SURVEY

- 2.1 Introduction
- 2.2 Battery Monitoring Systems in Electric Vehicles
- 2.3 CAN Communication in Automotive Systems
- 2.4 Embedded Systems for Real-Time Monitoring
- 2.5 IoT-Based Remote Battery Monitoring
- 2.6 Sensor Technologies for Battery Monitoring
- 2.7 Summary

3. AIM OF PROJECT

4.SCOPE AND OBJECTIVES

5. THEORETICAL DESCRIPTION OF PROJECT

- 5.1. Introduction
- 5.2. STM32F407 Microcontroller
- 5.3. ESP32 WROOM-32
- 5.4. INA219 Current and Voltage Sensor
- 5.5. NTC Thermistor
- 5.6. MQ135 Smoke Sensor
- 5.7. SN65HVD230 CAN Transceiver
- 5.8. Buck Converter
- 5.9. L293N Motor Drive
- 5.10. Transistor-Based Driver Circuit for Fan and Buzzer
- 5.11. CAN Communication Protocol
- 5.12. UART Communication
- 5.13. MQTT Protocol

6. SYSTEM DESIGN AND OPTIMIZATION

- 6.1. System Architecture
- 6.2. Hardware Design
- 6.3 Communication Architecture
- 6.4 Software Implementation
- 6.5 System Optimization

7. DEVELOPMENT TOOLS

- 7.1 STM32CubeIDE
- 7.2 FreeRTOS
- 7.3 Arduino IDE
- 7.4 MQTT Dashboard

8. VERIFICATION

9. CONCLUSION

10. FUTURE SCOPE

11. REFERENCES

ABSTRACT

The rapid adoption of Electric Vehicles (EVs) has increased the demand for reliable battery monitoring systems capable of ensuring safe and efficient battery operation. Battery parameters such as voltage, current, temperature, and smoke level must be continuously monitored to prevent hazardous conditions including overheating, excessive current draw, battery degradation, and fire risks. This project presents the design and implementation of a **Real-Time EV Battery Monitoring and Safety Protection System** using dual STM32F407 microcontrollers, CAN communication, and an MQTT-based IoT dashboard.

The proposed system consists of two STM32F407 microcontroller nodes. The first node acts as a **Sensor Acquisition Node**, collecting battery voltage and current data using the INA219 sensor through the I²C interface, battery temperature using an NTC thermistor, and smoke concentration using an MQ135 gas sensor through the ADC interface. Before transmission, the acquired sensor data undergoes validation, noise filtering, and fault checking to ensure reliable measurements. The processed data is then transmitted periodically to the second node through the CAN bus using SN65HVD230 CAN transceivers.

The second STM32F407 functions as the **Monitoring and Control Node**, where the received data is processed to calculate battery power, estimate the State of Charge (SOC), determine battery operating conditions, and identify abnormal operating situations such as under-voltage, over-voltage, over-current, excessive temperature, smoke detection, CAN communication failures, and sensor faults. Based on the detected conditions, appropriate safety actions such as activating a cooling fan and audible buzzer are performed. The processed battery information is simultaneously transmitted to an ESP32 module through UART communication, which publishes the data to an MQTT broker for real-time visualization on an IoT dashboard.

The developed system demonstrates reliable real-time battery parameter monitoring, distributed processing using CAN communication, and cloud-based remote monitoring through MQTT. The modular architecture enables easy scalability, simplifies maintenance, and provides an efficient platform for enhancing battery safety in electric vehicle applications.

Keywords: Electric Vehicle, Battery Monitoring System, STM32F407, CAN Bus, ESP32, MQTT, IoT, Battery Safety, INA219, MQ135.

INTRODUCTION

1.1 Introduction

Electric vehicles (EVs) have emerged as one of the most promising alternatives to conventional fuel-powered transportation due to their high energy efficiency, low emissions, and environmentally friendly operation. The battery pack serves as the primary energy source of an electric vehicle, making its continuous monitoring essential for ensuring safe, reliable, and efficient operation. Abnormal battery conditions such as excessive temperature, high current consumption, over-voltage, under-voltage, or smoke generation can significantly reduce battery life and may even lead to hazardous situations including thermal runaway and fire.

Modern battery monitoring systems continuously acquire battery parameters, process the collected information, and provide timely warnings whenever unsafe operating conditions are detected. In addition, remote monitoring through Internet of Things (IoT) technology enables users to supervise battery health from any location, allowing preventive maintenance and early fault detection.

This project presents the design and implementation of a **Real-Time EV Battery Monitoring and Safety Protection System** based on dual STM32F407 microcontrollers. The system employs distributed processing, where one STM32F407 functions as a sensor acquisition node while the second STM32F407 performs battery analysis, fault detection, and safety control. Communication between the two nodes is established using the Controller Area Network (CAN), a robust communication protocol widely used in automotive applications due to its reliability and fault tolerance.

Battery voltage and current are measured using the INA219 sensor through the I²C interface, while battery temperature and smoke concentration are monitored using an NTC thermistor and MQ135 gas sensor respectively through analog-to-digital conversion. The processed battery information is transmitted to an ESP32 module via UART communication and published to an MQTT-based IoT dashboard, enabling real-time visualization of battery parameters and fault conditions.

The developed system provides continuous battery monitoring, intelligent fault detection, safety protection through automatic fan and buzzer control, and remote monitoring capabilities. The modular architecture adopted in this project allows independent operation of sensing and monitoring units, improving system reliability and simplifying future expansion.

1.2 Problem Statement

The battery is one of the most critical and expensive components of an electric vehicle. Operating under abnormal conditions such as excessive temperature, high current, low battery voltage, or smoke generation can reduce battery life and compromise vehicle safety. Conventional battery monitoring systems often lack real-time fault detection, reliable communication, and remote monitoring capabilities.

Therefore, there is a need for a battery monitoring system that continuously monitors key battery parameters, detects abnormal conditions in real time, enables reliable communication between embedded controllers, and provides remote monitoring through IoT technology. Such a system enhances battery safety, improves reliability, and helps prevent failures before they become critical.

1.3 Objectives

The primary objectives of the proposed system are:

- To monitor battery voltage and current using the INA219 sensor.
- To measure battery temperature using an NTC thermistor.
- To detect smoke concentration using the MQ135 gas sensor.
- To establish reliable communication between two STM32F407 controllers using the CAN protocol.
- To implement real-time fault detection for abnormal battery operating conditions.
- To control safety devices such as the cooling fan and buzzer during fault conditions.
- To transmit battery data to an ESP32 using UART communication.
- To publish battery information to an MQTT-based IoT dashboard for remote monitoring.

1.4 Scope of the Project

The scope of this project includes the design and implementation of a real-time EV battery monitoring system capable of monitoring battery voltage, current, temperature, and smoke levels, detecting safety-related faults, and providing remote visualization through IoT technology. The system uses distributed STM32F407 controllers communicating via the CAN protocol, making it suitable for automotive communication networks.

The project is limited to battery parameter monitoring and safety indication. Features such as battery charging control, cell balancing, and advanced battery health prediction are beyond the scope of this work and can be considered for future enhancements.

1.5 Motivation

The increasing adoption of electric vehicles has highlighted the importance of ensuring battery safety and operational reliability. Continuous monitoring of battery parameters enables early detection of unsafe operating conditions, thereby preventing costly battery failures and improving user safety. Furthermore, integrating IoT technology allows users to monitor battery performance remotely, supporting predictive maintenance and efficient energy management. These requirements motivated the development of the proposed real-time battery monitoring and safety protection system.

LITERATURE SURVEY

2.1 Introduction

The rapid growth of electric vehicles (EVs) has increased the need for reliable battery monitoring systems to ensure safe and efficient operation. Continuous monitoring of battery parameters such as voltage, current, temperature, and smoke levels is essential to improve battery life and prevent failures. Advancements in embedded systems, CAN communication, and Internet of Things (IoT) technologies have enabled the development of intelligent battery monitoring solutions capable of real-time data acquisition, fault detection, and remote monitoring. This chapter reviews the key technologies used in battery monitoring systems and highlights the role of embedded controllers, CAN communication, and IoT in enhancing battery safety.

2.2 Battery Monitoring Systems in Electric Vehicles

Battery monitoring systems continuously measure battery parameters such as voltage, current, and temperature to detect abnormal operating conditions and improve battery safety. Modern systems employ distributed embedded controllers and communication networks to achieve reliable real-time monitoring, faster processing, and improved system scalability. These features support preventive maintenance and enhance the overall performance of electric vehicle battery systems.

2.3 CAN Communication in Automotive Systems

The **Controller Area Network (CAN)** protocol is a widely used communication standard in automotive electronics due to its reliability, fault tolerance, and efficient message transmission. It enables multiple electronic control units (ECUs) to communicate over a common two-wire bus while reducing wiring complexity. In battery monitoring applications, CAN ensures reliable exchange of sensor data and diagnostic information, making it well suited for real-time automotive systems.

2.4 Embedded Systems for Real-Time Monitoring

Modern battery monitoring systems use embedded microcontrollers to perform real-time data acquisition, processing, and fault detection. Microcontrollers such as the **STM32F407** provide multiple communication interfaces and sufficient processing capability for efficient battery monitoring. A distributed embedded architecture separates sensor acquisition from monitoring and control, improving system reliability, modularity, and ease of future expansion.

2.5 IoT-Based Remote Battery Monitoring

The Internet of Things (IoT) has transformed conventional battery monitoring by enabling remote access to real-time battery information through cloud platforms. Wireless communication modules such as the ESP32 allow embedded systems to publish sensor data using lightweight protocols like MQTT.

Cloud-based monitoring dashboards enable users to observe battery parameters, receive safety alerts, and analyse battery performance from any location with internet connectivity. Such systems improve operational efficiency and support preventive maintenance by allowing timely detection of abnormal battery conditions.

2.6 Sensor Technologies for Battery Monitoring

Accurate sensing of battery parameters is fundamental to reliable battery monitoring. Digital sensors such as the INA219 provide precise measurement of battery voltage and current using the I²C communication interface. Temperature monitoring is commonly performed using NTC thermistors due to their high sensitivity and low cost, while gas sensors such as the MQ135 can detect smoke or hazardous gases generated during battery failures.

The integration of multiple sensors enables comprehensive monitoring of battery operating conditions and improves the overall safety of electric vehicle battery systems.

2.7 Summary

The literature survey indicates that modern EV battery monitoring systems increasingly combine embedded controllers, CAN communication, intelligent sensing, and IoT technologies to achieve reliable real-time monitoring and safety protection. While many existing systems focus primarily on parameter measurement, integrating distributed processing, fault detection, and cloud-based visualization significantly enhances system reliability and user awareness.

The proposed project adopts these concepts by utilizing dual STM32F407 microcontrollers, CAN communication, multiple battery sensors, and an MQTT-based IoT dashboard to develop a modular and reliable real-time EV battery monitoring and safety protection system.

AIM OF PROJECT

The primary aim of this project is to design and implement a **Real-Time EV Battery Monitoring and Safety Protection System** using dual STM32F407 microcontrollers, CAN communication, and an MQTT-based IoT platform. The system continuously monitors critical battery parameters, including voltage, current, temperature, and smoke levels, to ensure safe and reliable battery operation.

The project aims to acquire sensor data in real time, transmit it reliably using the CAN protocol, detect abnormal operating conditions, and activate appropriate safety mechanisms such as a cooling fan and buzzer. Additionally, the system enables remote monitoring by publishing battery data to an MQTT-based IoT dashboard through an ESP32 module, allowing users to monitor battery health and receive fault notifications in real time.

The main objectives of the project are:

- To monitor battery voltage, current, temperature, and smoke levels in real time.
- To establish reliable communication between two STM32F407 controllers using the CAN protocol.
- To implement fault detection and safety protection for abnormal battery conditions.
- To provide remote battery monitoring through an MQTT-based IoT dashboard.
- To develop a modular and scalable embedded system suitable for EV battery monitoring applications

SCOPE AND OBJECTIVE

4.1 Scope of the Project

The scope of this project is to design and develop a **Real-Time EV Battery Monitoring and Safety Protection System** that continuously monitors key battery parameters, including voltage, current, temperature, and smoke levels. The system employs dual STM32F407 microcontrollers communicating over the CAN protocol to ensure reliable real-time data exchange. Processed battery information is transmitted to an ESP32 module and published to an MQTT-based IoT dashboard for remote monitoring. The project focuses on battery condition monitoring, fault detection, and safety indication through fan and buzzer control.

The project does not include battery charging control, cell balancing, or advanced battery health prediction techniques. These features can be incorporated in future developments.

4.2 Objectives of the Project

The main objectives of the proposed system are:

- To monitor battery voltage and current using the INA219 sensor.
- To measure battery temperature using an NTC thermistor.
- To detect smoke using the MQ135 gas sensor.
- To establish reliable communication between two STM32F407 controllers using the CAN protocol.
- To implement real-time fault detection for abnormal battery conditions.
- To control safety devices such as the cooling fan and buzzer during fault conditions.
- To transmit battery data to an ESP32 using UART communication.
- To provide remote battery monitoring through an MQTT-based IoT dashboard.

This format is concise, professional, and matches the style typically used in engineering project reports.

THEORETICAL DESCRIPTION OF PROJECT

5.1 Introduction

The **Real-Time EV Battery Monitoring and Safety Protection System** is designed to continuously monitor the operating condition of an electric vehicle battery by acquiring and processing critical parameters such as voltage, current, temperature, and smoke levels. The system utilizes two STM32F407 microcontrollers, where one functions as the **Sensor Acquisition Node** and the other as the **Monitoring and Control Node**. This distributed architecture improves system reliability and enables efficient processing of battery data.

Battery voltage and current are measured using the INA219 sensor through the I²C interface, while temperature and smoke levels are monitored using an NTC thermistor and MQ135 sensor through the STM32's Analog-to-Digital Converter (ADC). The acquired sensor data is validated and transmitted between the two STM32 controllers using the Controller Area Network (CAN) protocol through SN65HVD230 CAN transceivers.

The receiving STM32 processes the sensor data to calculate battery parameters, detect abnormal operating conditions, and activate safety mechanisms such as a cooling fan and buzzer whenever faults are detected. The processed information is then sent to an ESP32 module via UART communication, which publishes the data to an MQTT broker for real-time visualization on an IoT dashboard.

The combination of embedded processing, reliable CAN communication, and IoT connectivity provides a scalable, modular, and efficient solution for real-time EV battery monitoring and safety protection.

5.2 STM32F407 Microcontroller

Introduction

The STM32F407 Discovery Board is a high-performance development platform based on the STM32F407VGT6 microcontroller from STMicroelectronics. It is built around the ARM Cortex-M4 32-bit processor operating at a maximum clock frequency of 168 MHz. The microcontroller offers high computational capability, multiple communication interfaces, and low power consumption, making it suitable for real-time embedded and automotive applications.

In this project, two STM32F407 Discovery boards are used to implement a distributed battery monitoring architecture. One board functions as the Sensor Acquisition Node (STM1), while the second board acts as the Monitoring and Control Node (STM2). The two controllers communicate through the CAN protocol, enabling reliable and real-time exchange of battery data.

Key Features

- ARM Cortex-M4 32-bit processor with Floating Point Unit (FPU)
- Operating frequency up to 168 MHz
- 1 MB Flash memory
- 192 KB SRAM
- Multiple communication interfaces including CAN, UART, I²C, SPI, USB, and ADC
- Up to 140 GPIO pins
- Multiple 12-bit ADC channels
- Low power operation with high processing performance

Working Principle

The STM32F407 continuously acquires data from connected sensors and processes the received information using Embedded C firmware. In the transmitter node, sensor readings are validated, filtered, and transmitted to the receiver node over the CAN bus. The receiver node processes the received data, detects abnormal operating conditions, controls safety devices such as the fan and buzzer, and forwards battery information to the ESP32 module through UART for IoT monitoring.

Role in the Project

STM32F407 (Sensor Acquisition Node)

- Reads battery voltage and current from the INA219 via I²C.
- Acquires temperature data from the NTC thermistor using ADC.
- Reads smoke sensor values from the MQ135 using ADC.
- Validates sensor readings and filters noise.
- Packages sensor data into CAN frames.
- Transmits battery information to the receiver node every 100 ms.

STM32F407 (Monitoring and Control Node)

- Receives sensor data through the CAN interface.
- Calculates battery power and estimates the State of Charge (SOC).
- Detects faults such as over-voltage, under-voltage, over-current, over-temperature, smoke, and communication failures.
- Controls the cooling fan and buzzer based on detected fault conditions.
- Sends processed battery information to the ESP32 using UART communication for remote monitoring.

Advantages

- High processing speed suitable for real-time applications.
- Supports multiple communication protocols required in embedded systems.
- Reliable CAN controller for automotive communication.
- Large memory capacity for implementing complex monitoring algorithms.
- Modular architecture enables easy system expansion and maintenance.

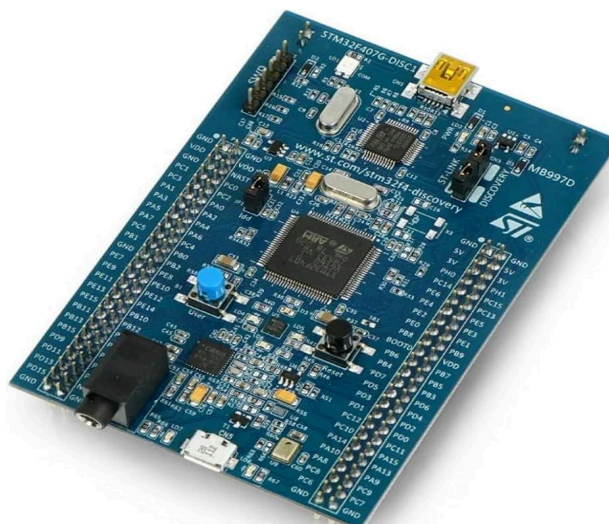


Figure 1: STM32F407VGT6

5.3 ESP32-WROOM-32

Introduction

The ESP32-WROOM-32 is a low-cost, high-performance Wi-Fi and Bluetooth-enabled microcontroller module developed by Espressif Systems. It is widely used in Internet of Things (IoT) applications due to its built-in wireless connectivity, low power consumption, and powerful dual-core processor.

In this project, the ESP32 serves as the IoT communication module. It receives processed battery parameters from the STM32F407 Monitoring and Control Node through UART communication and publishes the data to an MQTT broker over Wi-Fi. This enables real-time visualization of battery information on an IoT dashboard.

Key Features

- Dual-core Xtensa® 32-bit processor
- Operating frequency up to 240 MHz
- Built-in Wi-Fi (802.11 b/g/n)
- Bluetooth v4.2 and BLE support
- UART, SPI, I²C, PWM, and ADC interfaces
- Low power consumption
- Supports MQTT, HTTP, and TCP/IP protocols

Working Principle

The ESP32 receives battery information from STM32 through UART communication. It establishes a Wi-Fi connection and communicates with the MQTT broker to publish real-time sensor data. The published data is then displayed on the IoT dashboard, allowing users to monitor battery status remotely.

Role in the Project

- Receives battery parameters from STM32F407.
- Connects to the Wi-Fi network.
- Publishes sensor data using MQTT.
- Enables remote battery monitoring through the IoT dashboard.
- Displays real-time battery status and fault conditions.

Advantages

- Integrated Wi-Fi and Bluetooth.
- Fast data communication.
- Ideal for IoT applications.
- Easy integration with MQTT platforms.

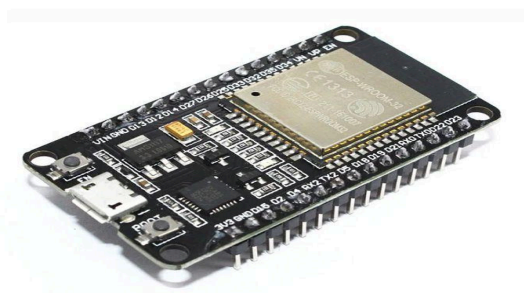


Figure 2: ESP32-WROOM-32 Module

5.4 INA219 Current and Voltage Sensor

Introduction

The INA219 is a high-precision digital sensor designed to measure both bus voltage and current through the I²C communication interface. It integrates a programmable gain amplifier (PGA) and a high-resolution Analog-to-Digital Converter (ADC), providing accurate electrical measurements while reducing external circuitry.

In this project, the INA219 continuously monitors the battery voltage and current, supplying essential information required for battery condition analysis and fault detection.

Key Features

- Measures voltage up to 26 V
- Bidirectional current measurement
- I²C communication interface
- 12-bit ADC resolution
- High measurement accuracy
- Low power consumption

Working Principle

The INA219 measures the voltage drop across a shunt resistor to calculate the battery current while simultaneously measuring the battery bus voltage. The measured values are digitally transmitted to the STM32F407 through the I²C interface for further processing.

Role in the Project

- Measures battery voltage.
- Measures charging/discharging current.
- Provides data for power calculation.
- Supports State of Charge (SOC) estimation.
- Assists in over-current and under-voltage detection.

Advantages

- High accuracy.
- Digital output through I²C.
- Simple interface.
- Low noise.

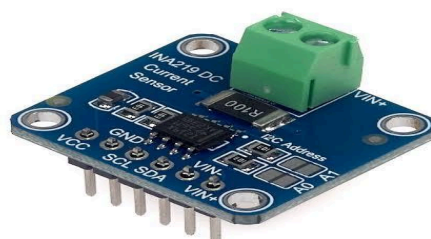


Figure 3: INA219 Voltage and Current Sensor

5.5 NTC Thermistor

Introduction

An NTC (Negative Temperature Coefficient) Thermistor is a temperature-sensitive resistor whose resistance decreases as temperature increases. Due to its high sensitivity, fast response, and low cost, it is widely used for temperature monitoring in electronic and automotive systems.

In this project, the NTC thermistor continuously monitors the battery temperature to detect overheating conditions and ensure safe battery operation.

Key Features

- High temperature sensitivity
- Fast response time
- Compact size
- Low cost
- Wide operating temperature range

Working Principle

The thermistor is connected as part of a voltage divider circuit. As temperature changes, its resistance changes, resulting in a varying output voltage. The STM32F407 reads this voltage using its ADC and converts it into temperature values through software calculations.

Role in the Project

- Monitor battery temperature.
- Detects overheating conditions.
- Activates the cooling fan.
- Triggers temperature warnings.
- Supports battery safety.

Advantages

- Simple construction.
- High accuracy.
- Fast response.
- Low power consumption.
- Reliable temperature measurement.



Figure 4: NTC Thermistor

5.6 MQ135 Smoke Sensor

Introduction

The MQ135 is a gas and smoke sensor capable of detecting harmful gases such as ammonia, carbon dioxide, benzene, and smoke particles. It is commonly used in air quality monitoring and fire detection systems.

In this project, the MQ135 sensor monitors the battery environment for smoke generation, which may indicate battery overheating or fire hazards.

Key Features

- Detects smoke and multiple gases
- Analog output
- High sensitivity
- Wide detection range
- Easy ADC interfacing

Working Principle

The sensing element changes its resistance in the presence of smoke or gases. This variation produces a corresponding analog voltage, which is converted into a digital value by the STM32 ADC. The software compares the measured value with predefined thresholds to determine the smoke status.

Role in the Project

- Detects smoke near the battery.
- Provides early fire warning.
- Generates emergency alerts.
- Activates the buzzer.
- Supports battery safety protection.

Advantages

- Low cost.
- Easy to interface.
- High sensitivity.
- Suitable for fire detection.
- Reliable operation.

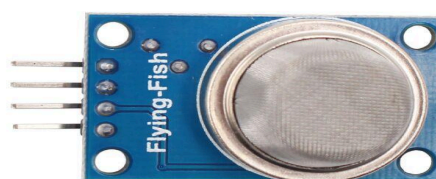


Figure 5: MQ135 Smoke Sensor

5.7 SN65HVD230 CAN Transceiver

Introduction

The **SN65HVD230** is a high-speed Controller Area Network (CAN) transceiver developed by Texas Instruments. It provides the physical interface between the CAN controller of the STM32F407 microcontroller and the CAN bus by converting digital CAN transmit (TX) and receive (RX) signals into differential signals. This ensures reliable communication even in electrically noisy environments.

In this project, two SN65HVD230 transceivers are used to establish CAN communication between the **Sensor Acquisition Node (STM32F407 - STM1)** and the **Monitoring and Control Node (STM32F407 - STM2)** for transmitting battery parameters and diagnostic information in real time.

Key Features

- Supports CAN 2.0A and CAN 2.0B standards.
- Data communication up to **1 Mbps**.
- Compatible with **3.3 V** STM32F407 microcontrollers.
- Differential signaling for high noise immunity.
- Low power consumption with built-in protection features.

Working Principle

The STM32F407 sends CAN messages to the SN65HVD230 through the **CAN_TX** pin. The transceiver converts these digital signals into differential signals on the **CANH** and **CANL** lines for transmission over the CAN bus. At the receiving node, the transceiver converts the differential signals back into digital data and forwards them to the STM32F407 through the **CAN_RX** pin.

Role in the Project

- Enables CAN communication between the two STM32F407 controllers.
- Transfers battery voltage, current, temperature, smoke data, and fault information.
- Supports reliable real-time monitoring in automotive environments.

Advantages

- High communication reliability.
- Excellent noise immunity.
- Compatible with 3.3 V microcontrollers.
- Suitable for automotive and industrial applications.



Figure 6: SN65HVD230 CAN Transceiver

5.8 Buck Converter

Introduction

A Buck Converter is a DC-DC step-down converter that efficiently reduces a higher input voltage to a lower regulated output voltage. It operates using high-frequency switching techniques, making it more efficient than linear voltage regulators. Buck converters are widely used in embedded and automotive systems to provide stable power to electronic components.

In this project, the buck converter is used to step down the 12 V battery voltage to the required operating voltage for the motor driver and other electronic components, ensuring stable and efficient power distribution.

Key Features

- High conversion efficiency
- Stable regulated output voltage
- Low heat generation
- Compact and lightweight design
- Wide input voltage range

Working Principle

The buck converter rapidly switches the input voltage using a transistor and stores energy in an inductor. The output voltage is regulated by controlling the switching duty cycle, resulting in a lower and stable DC voltage suitable for powering connected devices.

Role in the Project

- Converts battery voltage to the required operating level.
- Supplies stable power to the motor driver.
- Improves overall power efficiency.
- Protects electronic components from voltage fluctuations.

Advantages

- High efficiency compared to linear regulators.
- Low power loss.
- Stable voltage output.
- Compact design.
- Suitable for battery-powered applications.

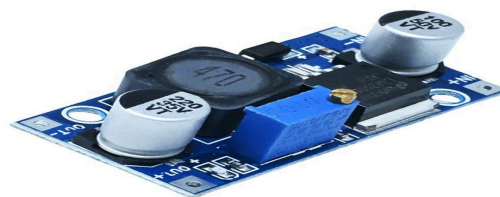


Figure 7: Buck Converter Module

5.9 L293N Motor Driver

Introduction

The L293N is a dual H-Bridge motor driver IC designed to control the direction and speed of DC motors. It provides sufficient current to drive motors while protecting the microcontroller from high-current loads.

In this project, the L293N motor driver is used to control the 12 V DC motor, which represents the motor load of the electric vehicle battery system during testing.

Key Features

- Dual H-Bridge driver
- Supports two DC motors
- Motor supply voltage up to 36 V
- Output current up to 600 mA per channel
- Built-in protection diodes
- TTL compatible inputs

Working Principle

The STM32F407 generates digital control signals that are applied to the input pins of the L293N. Based on these signals, the motor driver controls the direction and operation of the DC motor by switching the current flow through the motor terminals.

Role in the Project

- Drives the 12 V DC motor.
- Simulates battery load conditions.
- Provides isolation between the STM32 and motor.
- Enables safe motor operation.

Advantages

- Easy to interface.
- Reliable motor control.
- Built-in protection.
- Supports bidirectional motor operation.
- Widely used in embedded applications.

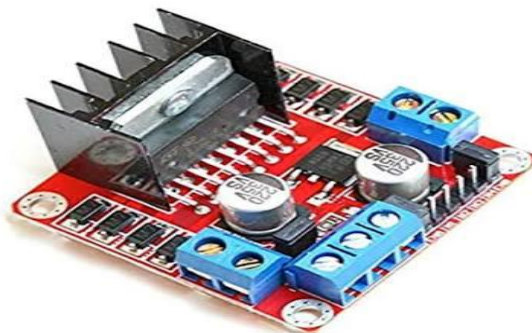


Figure 8: L293N Motor Driver Module

5.10 Transistor-Based Driver Circuit for Fan and Buzzer

Introduction

The Fan and Buzzer Driver Circuit is designed to interface high-current output devices with the STM32F407 microcontroller. Since the GPIO pins of the STM32 cannot directly drive devices such as a DC fan or buzzer, a transistor-based driver circuit is used. Both the fan and buzzer employ identical driver circuits consisting of an NPN transistor, a 320 Ω base resistor, and a Zener diode for voltage protection.

In this project, separate driver circuits are implemented for the cooling fan and buzzer. The fan provides automatic battery cooling during high-temperature conditions, while the buzzer generates audible alerts whenever a fault is detected.

Key Features

- Transistor-based switching circuit
- Supports higher current loads
- Protects STM32 GPIO pins
- Fast and reliable switching
- Simple and cost-effective design
- Zener diode for voltage protection

Working Principle

The STM32F407 outputs a digital control signal to the base of the NPN transistor through a 320 Ω resistor. When the GPIO pin is driven HIGH, the transistor saturates and acts as an electronic switch, allowing current to flow from the power supply to the connected device (fan or buzzer). When the GPIO output is LOW, the transistor switches OFF, interrupting the current flow and turning the device OFF. The Zener diode protects the circuit from voltage transients and ensures safe operation of the switching components.

Role in the Project

Fan Driver

- Activates the cooling fan when the battery temperature exceeds the predefined threshold.
- Automatically switches OFF the fan once the temperature returns to a safe operating range.

Buzzer Driver

- Generates audible alerts during fault conditions.
- Indicates events such as over-voltage, under-voltage, over-current, over-temperature, smoke detection, and communication failures.

Advantages

- Simple and reliable switching mechanism.
- Protects the STM32 microcontroller from high-current loads.
- Fast response during fault conditions.
- Improves overall system safety and reliability.

5.11 CAN Communication Protocol

Introduction

The Controller Area Network (CAN) is a robust serial communication protocol developed by Robert Bosch GmbH for reliable communication between Electronic Control Units (ECUs) in automotive and industrial applications. It enables multiple devices to communicate over a common two-wire bus without requiring a host controller, reducing wiring complexity and improving system reliability.

In this project, the CAN protocol is used for communication between the Sensor Acquisition Node (STM32F407 - STM1) and the Monitoring and Control Node (STM32F407 - STM2) through SN65HVD230 CAN transceivers. It ensures reliable and real-time transmission of battery parameters and diagnostic information.

Key Features

- Multi-master communication architecture.
- High-speed data transmission up to 1 Mbps.
- Differential signaling using CANH and CANL for high noise immunity.
- Built-in error detection and error handling mechanisms.
- Message prioritization using unique message identifiers.
- Reliable communication suitable for automotive applications.

Working Principle

The STM32F407 Sensor Acquisition Node continuously reads battery voltage, current, temperature, and smoke sensor data. After validating and filtering the sensor readings, the data is packed into a CAN data frame and transmitted every 100 ms.

The Monitoring and Control Node receives the CAN message, extracts the sensor values, and processes them to calculate battery parameters and detect abnormal operating conditions. Based on the received information, the system activates the cooling fan or buzzer when necessary and forwards the processed data to the ESP32 module for IoT monitoring.

Role in the Project

- Transfers battery sensor data between the two STM32F407 controllers.
- Ensures reliable real-time communication.
- Reduces communication wiring between embedded nodes.
- Transmits both sensor values and diagnostic fault information.
- Supports distributed processing in the battery monitoring system.

Advantages

- High communication reliability.
- Excellent noise immunity for automotive environments.
- Low wiring complexity.
- Fast and deterministic message transmission.
- Supports real-time embedded applications.

5.12 UART Communication

Introduction

Universal Asynchronous Receiver-Transmitter (UART) is a widely used serial communication protocol that enables data exchange between two devices without requiring a shared clock signal. It is simple, reliable, and commonly used in embedded systems for short-distance communication.

In this project, UART communication is established between the STM32F407 Monitoring and Control Node (STM2) and the ESP32 module. After processing the received battery data, STM2 transmits the battery parameters and fault status to the ESP32, which then forwards the information to the MQTT-based IoT dashboard.

Key Features

- Asynchronous serial communication.
- Simple two-wire interface (TX and RX).
- Full-duplex communication.
- Configurable baud rate.
- Low hardware complexity.
- Reliable communication for embedded systems.

Working Principle

The STM32F407 Monitoring and Control Node formats the processed battery information, including voltage, current, temperature, smoke level, State of Charge (SOC), and fault status, into serial data packets. These packets are transmitted through the UART TX pin and received by the ESP32 through its RX pin. The ESP32 decodes the received data and publishes it to the MQTT broker for remote monitoring.

Role in the Project

- Transfers processed battery information from STM32F407 to ESP32.
- Provides reliable communication between the embedded controller and IoT module.
- Enables real-time transmission of battery parameters.
- Supports remote monitoring through MQTT.

Advantages

- Simple and easy to implement.
- Low communication overhead.
- Reliable data transfer over short distances.
- Requires minimal hardware connections.
- Widely supported by embedded microcontrollers.

5.13 MQTT Protocol

Introduction

Message Queuing Telemetry Transport (MQTT) is a lightweight messaging protocol designed for Internet of Things (IoT) applications. It follows a publish-subscribe communication model, where devices publish data to an MQTT broker, and subscribed clients receive the data in real time. Due to its low bandwidth requirement and high efficiency, MQTT is widely used in remote monitoring applications.

In this project, the ESP32 publishes battery parameters received from the STM32F407 to an MQTT broker over Wi-Fi. The IoT dashboard subscribes to the published topics and displays real-time battery information to the user.

Key Features

- Lightweight publish-subscribe protocol.
- Low network bandwidth usage.
- Fast and efficient communication.
- Reliable message delivery.
- Suitable for IoT and remote monitoring applications.

Working Principle

After receiving battery data from the STM32F407 via UART, the ESP32 connects to a Wi-Fi network and establishes communication with the MQTT broker. It publishes battery parameters such as voltage, current, temperature, smoke level, SOC, and fault status to predefined MQTT topics. The IoT dashboard subscribes to these topics and updates the displayed information in real time.

Role in the Project

- Enables wireless transmission of battery data.
- Publishes real-time battery parameters to the MQTT broker.
- Supports remote monitoring through the IoT dashboard.
- Allows users to monitor battery health from any location.

Advantages

- Lightweight and efficient communication protocol.
- Low power and bandwidth consumption.
- Supports real-time monitoring.
- Highly scalable for IoT applications.
- Reliable communication over wireless networks.

SYSTEM DESIGN AND OPTIMIZATION

The proposed **Real-Time EV Battery Monitoring and Safety Protection System** is designed using a distributed embedded architecture to continuously monitor the operating condition of an electric vehicle battery. The system consists of two STM32F407 microcontrollers, multiple sensors, an ESP32 Wi-Fi module, and an MQTT-based IoT dashboard for remote monitoring.

The first STM32F407 acts as the **Sensor Acquisition Node (STM1)**. It acquires battery voltage and current from the INA219 sensor through the I²C interface, battery temperature from the NTC thermistor, and smoke level from the MQ135 sensor using ADC channels. The acquired sensor data is validated and transmitted to the second STM32F407 through the CAN bus using SN65HVD230 CAN transceivers.

The second STM32F407 functions as the **Monitoring and Control Node (STM2)**. It receives the sensor data, performs battery parameter calculations, detects fault conditions such as under-voltage, over-voltage, over-current, over-temperature, smoke detection, and communication failures, and activates safety devices including the cooling fan and buzzer whenever required.

The processed battery information is then transmitted to the ESP32 via UART communication. The ESP32 publishes the data to an MQTT broker over Wi-Fi, allowing real-time visualization of battery parameters and fault status on an IoT dashboard.

The distributed design separates sensor acquisition from monitoring and control, improving system reliability, reducing processing load, and enabling modular system development. This architecture also allows future expansion by incorporating additional sensors, battery management functions, or cloud-based analytics without significant hardware modifications.

6.1 System Architecture

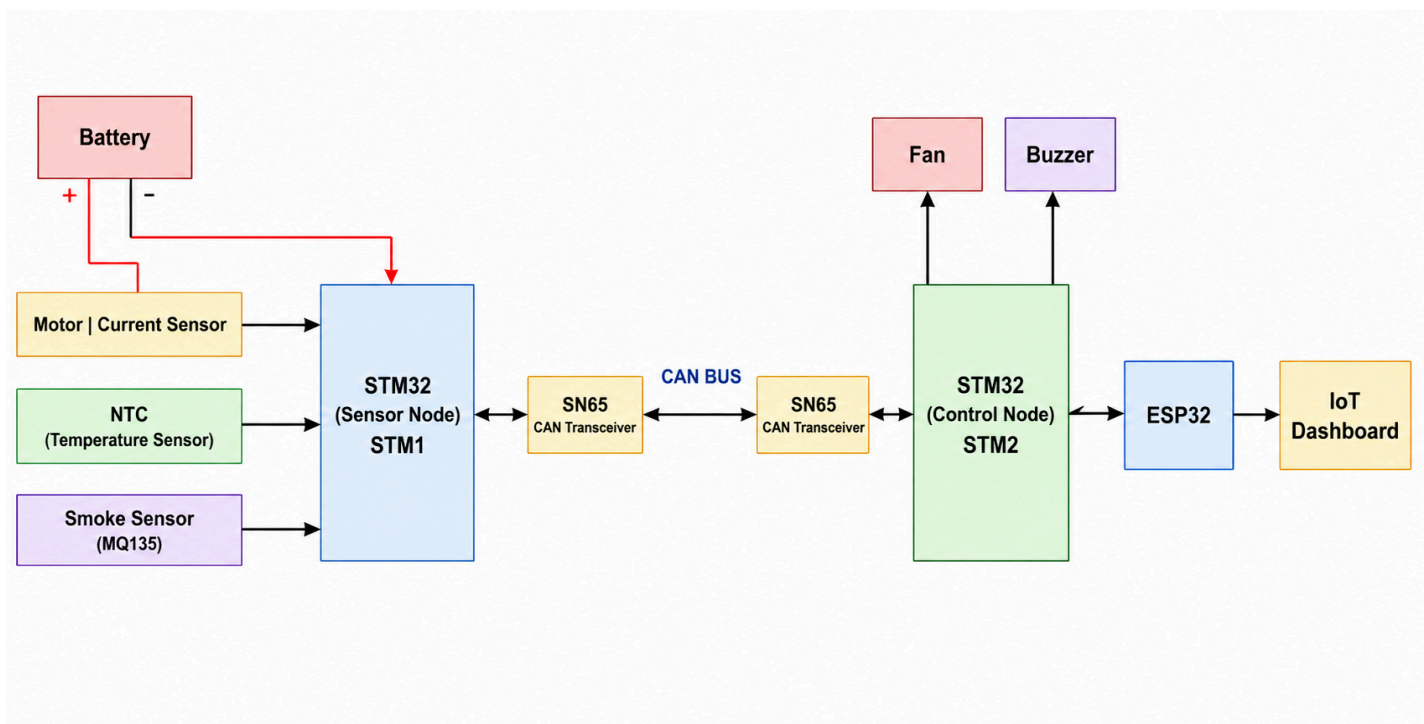


Figure 9: Detailed Block Diagram

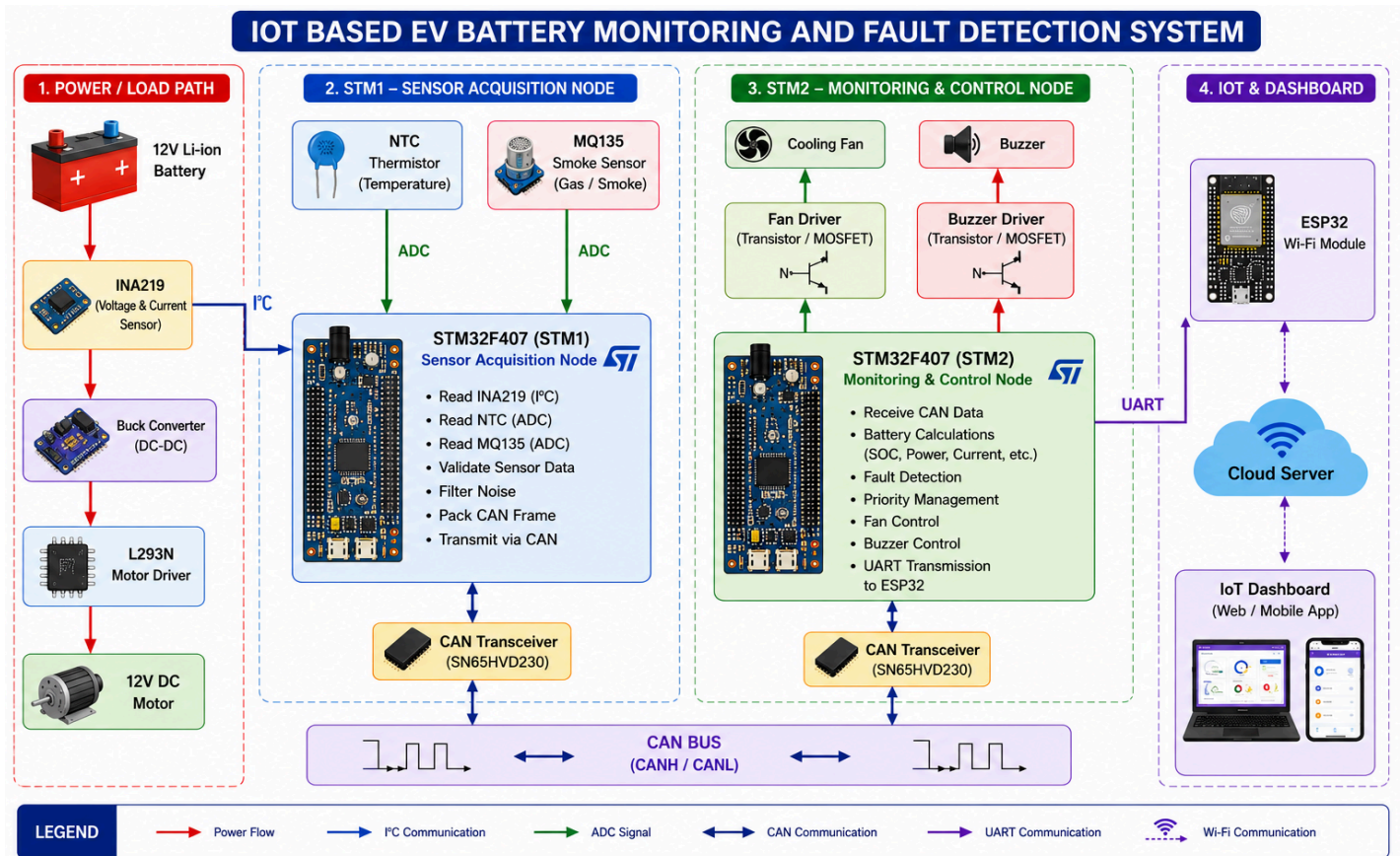


Figure 10: Detailed Block Diagram

6.2 Hardware Design

6.2.1 Sensor Acquisition Node (STM32F407 - STM1)

- STM1 acts as the transmitter node.
- INA219 measures battery voltage and current through I²C.
- NTC Thermistor measures battery temperature using ADC.
- MQ135 detects smoke using ADC.
- Sensor values are validated before transmission.
- Data is transmitted to STM2 using CAN through SN65HVD230.

6.2.2 Monitoring and Control Node (STM32F407 - STM2)

- STM2 acts as the receiver node.
- Receives CAN data.
- Performs battery calculations.
- Detects faults.
- Controls fan and buzzer.
- Sends processed data to ESP32 through UART.

6.2.3 IoT Module (ESP32)

- Receives data from STM2.
- Connects to Wi-Fi.
- Publishes data using MQTT.
- Dashboard displays battery parameters.

6.2.4 Power Supply and Load

- 12V battery.
- Buck converter.
- L293N motor driver.
- 12V DC motor as load.
- Fan and buzzer powered through transistor driver circuits.

6.3 Communication Architecture

- CAN Bus Communication (STM1 ↔ STM2)
- UART Communication (STM2 ↔ ESP32)
- MQTT Communication (ESP32 ↔ Cloud Dashboard)
- I²C Communication (STM1 ↔ INA219)

6.4 Software Implementation

The software for the proposed system is divided into three functional modules: the Sensor Acquisition Node (STM1), the Monitoring and Control Node (STM2), and the ESP32 IoT module.

The STM1 firmware continuously acquires battery voltage, current, temperature, and smoke sensor readings. The acquired values are validated to eliminate invalid measurements, filtered to reduce noise, and transmitted periodically to STM2 through the CAN bus.

The STM2 firmware receives the sensor data and performs battery parameter calculations and fault detection. Based on predefined threshold values, it controls the cooling fan and buzzer whenever abnormal operating conditions are detected. The processed battery information is then transmitted to the ESP32 through UART communication.

The ESP32 establishes a Wi-Fi connection and publishes the received battery data to an MQTT broker, enabling real-time visualization on the IoT dashboard.

6.4.1 STM1 Firmware

- Initializes ADC, I²C, CAN.
- Read sensors.
- Validates sensor values.
- Filter noise.
- Packs CAN message.
- Sends every 100 ms.

6.4.2 STM2 Firmware

- Initializes CAN and UART.
- Receives CAN frame.
- Calculates:
 - Battery Power
 - SOC
 - Load
 - Battery Status
- Detects faults.
- Control fan.
- Controls buzzer.
- Send data to ESP32.

6.4.3 ESP32 Firmware

- Initializes UART and Wi-Fi.
- Receives battery data.
- Connects to MQTT broker.
- Publishes battery parameters.
- Dashboard updates automatically.

6.5 System Optimization

Although advanced optimization techniques were not implemented, several methods were incorporated to improve the reliability and efficiency of the system.

- Sensor values are validated before transmission to prevent invalid measurements.
- Multiple sensor samples are averaged to reduce noise and improve measurement stability.
- Battery data is transmitted over the CAN bus at regular intervals (100 ms), ensuring reliable real-time communication.
- A fault-priority mechanism ensures that the most critical battery fault is handled first.
- Temperature hysteresis is implemented for fan control to prevent frequent ON/OFF switching near the threshold temperature.

These techniques improve communication reliability, reduce unnecessary processing, and enhance the overall performance of the battery monitoring system.

DEVELOPMENT TOOLS

7.1 STM32CubeIDE

STM32CubeIDE is the primary Integrated Development Environment (IDE) used for developing firmware for the STM32F407 microcontrollers. It provides code editing, debugging, peripheral configuration, and firmware programming in a single platform. In this project, STM32CubeIDE was used to develop and debug the firmware for both the Sensor Acquisition Node (STM1) and the Monitoring and Control Node (STM2).

7.2 FreeRTOS

FreeRTOS is an open-source Real-Time Operating System (RTOS) designed for embedded systems. It enables efficient task scheduling, resource management, and real-time execution of multiple operations.

In this project, FreeRTOS was used to manage concurrent tasks such as sensor data acquisition, CAN communication, fault detection, UART communication, and MQTT data transmission. The RTOS scheduler ensured timely execution of these tasks, improving system responsiveness and overall reliability.

Features of FreeRTOS:

- Priority-based task scheduling
- Real-time task execution
- Efficient CPU utilization
- Low memory footprint
- Inter-task synchronization and communication

7.3 Arduino IDE

Arduino IDE was used to develop and upload firmware to the ESP32 module. The ESP32 receives battery data from STM32F407 via UART communication and publishes the information to the MQTT broker over Wi-Fi for remote monitoring.

7.4 MQTT Dashboard

The MQTT Dashboard is an IoT visualization platform used to monitor battery parameters remotely. It subscribes to MQTT topics published by the ESP32 and displays real-time battery information such as voltage, current, temperature, smoke level, battery status, and fault conditions.

VERIFICATION

8.1 Introduction

Verification is the process of testing the developed system to ensure that all hardware modules, communication interfaces, and software functionalities operate as intended. In the proposed Real-Time EV Battery Monitoring and Safety Protection System, each subsystem was tested individually before integrating the complete system. The verification process confirmed correct sensor data acquisition, reliable communication between embedded controllers, fault detection, safety device operation, and successful transmission of battery data to the MQTT dashboard.

8.2 Hardware Verification

The hardware components were tested individually to verify proper operation before system integration.

Component	Verification Performed	Status
STM32F407 (STM1)	Sensor acquisition and CAN transmission	Verified
STM32F407 (STM2)	CAN reception and fault detection	Verified
INA219	Voltage and current measurement	Verified
NTC Thermistor	Temperature measurement	Verified
MQ135	Smoke detection	Verified
SN65HVD230	CAN communication	Verified
ESP32	UART and MQTT communication	Verified
Fan Driver Circuit	Fan switching during high temperature	Verified
Buzzer Driver Circuit	Audible alert during faults	Verified

8.3 Communication Verification

The communication interfaces used in the system were verified individually.

Communication	Purpose	Verification Result
CAN Bus	STM1 ↔ STM2	Sensor data transmitted successfully
UART	STM2 ↔ ESP32	Battery data received successfully
MQTT	ESP32 ↔ Dashboard	Real-time data displayed successfully
I ² C	STM1 ↔ INA219	Voltage and current measured correctly

8.4 Functional Verification

The implemented software functions were tested under different operating conditions.

Function	Expected Result	Status
Voltage Measurement	Battery voltage displayed correctly	Verified
Current Measurement	Current measured correctly	Verified
Temperature Monitoring	Temperature updated correctly	Verified
Smoke Detection	Smoke level detected	Verified
CAN Communication	Data exchanged without errors	Verified

Fan Control	Fan ON above threshold temperature	Verified
Buzzer Control	Alarm generated during fault	Verified
MQTT Dashboard	Live battery data displayed	Verified

8.5 Fault Verification

Different fault conditions were simulated to verify the response of the monitoring system.

Fault Condition	Expected Action	Observed Result
Under Voltage	Warning generated	Verified
Over Voltage	Fault indication	Verified
Over Current	Buzzer activated	Verified
High Temperature	Fan activated	Verified
Smoke Detection	Continuous buzzer alert	Verified
CAN Communication Failure	Communication fault detected	Verified

8.6 Verification Summary

The verification process confirmed that all hardware modules, communication interfaces, and software functionalities operated as expected. Sensor data was acquired successfully, CAN communication remained reliable, fault conditions were detected correctly, and safety devices responded according to the implemented logic. The ESP32 successfully transmitted battery information to the MQTT dashboard, enabling real-time remote monitoring of the battery system.

CONCLUSION

The Real-Time EV Battery Monitoring and Safety Protection System was successfully designed and implemented using two STM32F407 microcontrollers, multiple sensors, CAN communication, and an MQTT-based IoT platform. The system continuously monitors critical battery parameters such as voltage, current, temperature, and smoke level to ensure safe battery operation.

A distributed architecture was adopted in which one STM32 controller performs sensor acquisition while the second controller handles battery monitoring, fault detection, and safety control. Reliable communication between the controllers was achieved using the CAN protocol through SN65HVD230 transceivers. The system successfully detects abnormal operating conditions such as under-voltage, over-voltage, over-current, over-temperature, smoke detection, and communication failures, and responds by activating the cooling fan and buzzer.

Furthermore, the integration of the ESP32 with an MQTT-based IoT dashboard enables real-time remote monitoring of battery parameters, improving accessibility and user awareness. The modular design, reliable communication, and real-time safety mechanisms make the proposed system suitable for electric vehicle battery monitoring applications and provide a strong foundation for future Battery Management System (BMS) development.

FUTURE SCOPE

The proposed system can be further enhanced by incorporating additional Battery Management System (BMS) features to improve battery safety, performance, and reliability. Some possible future improvements include:

- Integration of **cell balancing** techniques to improve battery life and charging efficiency.
- Implementation of **State of Health (SOH)** estimation for battery degradation analysis.
- Addition of **Coulomb Counting** for more accurate State of Charge (SOC) estimation.
- Integration of **GPS** for vehicle location tracking during emergency conditions.
- Development of a **mobile application** for remote battery monitoring and instant notifications.
- Implementation of **cloud-based data analytics** for predictive maintenance and fault prediction.
- Support for monitoring larger battery packs with multiple sensor nodes.
- Integration of machine learning algorithms for intelligent fault diagnosis and battery health prediction.

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